

VUKO MNGOMEZULU

Computer Science & Mathematics

vukokevin@gmail.com | <https://github.com/vukomngomezulu> | [LinkedIn: linkedin.com/in/vuko-mngomezulu-18a097273/](https://www.linkedin.com/in/vuko-mngomezulu-18a097273/)

PROFESSIONAL SUMMARY

Final-year Computer Science and Mathematics student with experience building AI-enabled applications, data pipelines, backend systems and automation-focused software projects. Experienced in Python, SQL, FastAPI, Docker and cloud technologies, with practical exposure to Retrieval-Augmented Generation (RAG), LLM applications, document processing, data engineering and AI-assisted workflows. A curious, hands-on builder who enjoys taking ambiguous problems from idea to working prototype. Projects include a local RAG document intelligence assistant, a real-time fraud detection and an agricultural marketplace platform. Particularly interested in AI engineering, intelligent automation, data systems and building practical AI products that solve real-world problems.

CORE SKILLS

Programming & Development: Python, SQL, FastAPI, REST APIs, React, Git/GitHub

AI & Intelligent Systems: Generative AI & LLM Applications, Retrieval-Augmented Generation (RAG), AI-assisted Workflows, Document Intelligence, Embeddings & Vector Search

Data: MySQL, SQL, PostgreSQL, relational database concepts, data modelling fundamentals, structured data, CRUD operations, data retrieval, Apache Kafka

Database Operations: Database health and reliability fundamentals, query troubleshooting, and deployments.

Cloud & Tools: Microsoft Azure fundamentals (AZ-900), Docker

EDUCATION

University of Zululand — Bachelor of Science in Computer Science and Mathematics | Final Year 2026

Academic performance: Multiple distinctions in Mathematics and Computer Science modules.

CERTIFICATION

Microsoft Certified: Azure Fundamentals (AZ-900)

RELEVANT PROJECT EXPERIENCE

Local RAG Document Intelligence Assistant

Python | FastAPI | React | ChromaDB | Ollama | SQLite | Docker

- Built a local AI-powered document question-answering application using Retrieval-Augmented Generation (RAG) to retrieve relevant information from user documents before generating responses.
- Designed a document-processing pipeline using PDF and DOCX extraction, chunking, embeddings and vector search.
- Integrated local embedding and LLM models through Ollama, enabling AI functionality without relying on paid cloud AI APIs.
- Developed a FastAPI backend and React frontend for document ingestion, querying and conversational interaction.
- Used ChromaDB for persistent vector storage and SQLite for conversation data.
- Containerised application components using Docker to improve reproducibility and deployment consistency.
- Explored techniques for improving retrieval quality, document context and AI-generated responses.

Real-Time Fraud Detection & ETL Platform

Python | Apache Kafka | PostgreSQL | Flask | Docker

- Developed an event-driven data pipeline for processing simulated financial transactions in real time.
- Built ETL workflows to generate, ingest, transform and analyse transaction data using Apache Kafka and PostgreSQL.
- Implemented rule-based risk detection to identify suspicious transaction patterns and generate explainable risk indicators.
- Developed a Flask dashboard to visualise transaction and fraud-detection results.
- Containerised the Kafka, database and application components using Docker.

Applied data engineering concepts including streaming, transformation, validation and persistent storage.